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OPTICAL ELEMENT DRIVE MECHANISM

Abstract

An optical element drive mechanism is provided. The optical element drive mechanism has a central axis. The optical element drive mechanism includes an immovable part, a first movable part, and a first drive assembly. The immovable part includes a bottom. The first movable part is connected to a first optical element. The first movable part is movable relative to the bottom. The first drive assembly drives the first movable part to move relative to the bottom.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/562,767, filed on Mar. 8, 2024, the entirety of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to an optical element drive mechanism, and, in particular, to an optical element drive mechanism including a plurality of circuit elements.

Description of the Related Art

[0003] As technology has developed, many electronic devices (such as tablet computers and smartphones) may be used for capturing images and recording video. The optical element and the optical element drive mechanism in the electronic device allow the user to use the electronic device to capture images and record video. When the electronic device is being used, shock or vibration may occur, and this may cause the images or video to come out blurry. Therefore, the demand for higher quality images and video is increasing.

BRIEF SUMMARY OF THE INVENTION

[0004] An optical element drive mechanism having a central axis is provided. The optical element drive mechanism includes an immovable part, a first movable part, and a first drive assembly. The immovable part includes a bottom. The first movable part is connected to a first optical element. The first movable part is movable relative to the bottom. The first drive assembly drives the first movable part to move relative to the bottom.

[0005] In some embodiments, the optical element drive mechanism further includes a first-side circuit element electrically connected to the first drive assembly. The bottom includes an embedded circuit, and the first-side circuit element includes a first portion of the first-side circuit element having a plate-like structure, and the first portion of the first-side circuit element is disposed on the bottom. The first portion of the first-side circuit element includes an electrical connection portion of the first-side circuit element, and the first portion of the first-side circuit element is electrically connected to the embedded circuit of the bottom via the electrical connection portion of the first-side circuit element. The bottom includes a bottom first surface facing the first portion of the first-side circuit element, and when viewed along a direction that is perpendicular to the central axis, the bottom first surface is located between a center of the first movable part and the first portion of the first-side circuit element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The present disclosure can be more fully understood by reading the detailed description and examples with references made to the accompanying drawings. It should be noted that various features may be not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or decreased for clarity of discussion, and the various features may be drawn schematically.

[0007] FIG. 1 is a perspective view of the optical element drive mechanism in accordance with some embodiments.

[0008] FIG. 2 is an exploded view of the optical element drive mechanism in accordance with some embodiments.

[0009] FIG. 3 is a schematic view of the casing in accordance with some embodiments.

[0010] FIG. 4 is a schematic view of the mounting element in accordance with some embodiments.

[0011] FIG. 5 is a schematic view of the bottom in accordance with some embodiments.

[0012] FIG. 6 is a perspective view of the first movable part in accordance with some embodiments.

[0013] FIG. 7 is a perspective view of the second movable part in accordance with some embodiments.

[0014] FIG. 8A, FIG. 8B, and FIG. 8C are schematic views illustrating the operation of the second drive assembly in driving the second movable part in accordance with some embodiments.

[0015] FIG. 9 is a schematic view of the first-side circuit element and the third-side circuit element in accordance with some embodiments.

[0016] FIG. 10 is a schematic view of the external connection circuit element in accordance with some embodiments.

[0017] FIG. 11 and FIG. 12 are schematic views of the bottom and the circuit assembly from different perspectives in accordance with some embodiments.

[0018] FIG. 13, FIG. 14, and FIG. 15 are perspective views of the optical element drive mechanism with the casing omitted from different perspectives in accordance with some embodiments.

[0019] FIG. 16 is a top view of the optical element drive mechanism with the casing omitted in accordance with some embodiments.

[0020] FIG. 17 and FIG. 18 are perspective views of the optical element drive mechanism with the casing, the mounting element, and the second movable part omitted in accordance with some embodiments.

[0021] FIG. 19 is a cross-sectional view of the optical element drive mechanism with the casing omitted in accordance with some embodiments.

DETAILED DESCRIPTION OF THE INVENTION

[0022] The following description provides different embodiments, or examples, for implementing different features of the present disclosure. In addition, spatially relative terms may be used in the following description to describe the arrangements of various features. These spatially relative terms are for ease of describing the positional relationship between one feature and another feature as illustrated in the drawings. These spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation illustrated in the drawings. The device may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative terms used in the following description may likewise be interpreted accordingly. For example, if a device of the drawings is flipped upside down, a feature that is “above” will become a feature that is “below”.

[0023] For example, the formation of a first feature “on,” “over,” or “under” a second feature in the following description may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first feature and the second feature, such that the first feature and the second feature are not in direct contact.

[0024] In the following description, the terms “including”, “comprising”, “having”, and the like should be interpreted as meaning “including but not limited to . . .”. Therefore, when the terms “including”, “comprising”, “having”, and the like are used, the presence of corresponding features, regions, steps, operations and/or elements is specified, and without excluding the presence of other features, regions, steps, operations and/or elements.

[0025] Please refer to FIG. 1 and FIG. 2 to understand an optical element drive mechanism **100**.

FIG. 1 is a perspective view of the optical element drive mechanism **100** in accordance with some embodiments. FIG. 2 is an exploded view of the optical element drive mechanism **100** in accordance with some embodiments.

[0026] The optical element drive mechanism **100** has a central axis C. The central axis C is an imaginary axis passing through the center of the optical element drive mechanism **100**. When viewed along the central axis C, the optical element drive mechanism **100** is polygonal. For ease of

explanation, the four sides of the optical element drive mechanism **100** are defined as a first side **1001**, a second side **1002**, a third side **1003**, and a fourth side **1004**. The first side **1001** is opposite to the third side **1003**, and the second side **1002** is opposite to the fourth side **1004**. The first side **1001** is substantially parallel with the third side **1003**, and the second side **1002** is substantially parallel with the fourth side **1004**. The first side **1001**, the second side **1002**, the third side **1003**, and the fourth side **1004** are substantially perpendicular to the central axis C.

[0027] When viewed along the central axis C, the first side **1001** and the third side **1003** are parallel with a first axis A1 and extend along the first axis A1. When viewed along the central axis C, the second side **1002** and the fourth side **1004** are parallel with a second axis A2 and extend along the second axis A2. The first axis A1 and the second axis A2 are substantially perpendicular to the central axis C.

[0028] In addition, the four corners of the optical element drive mechanism **100** are defined as a first corner **2001**, a second corner **2002**, a third corner **2003**, and a fourth corner **2004**. The first corner **2001** is located between the first side **1001** and the fourth side **1004**. The second corner **2002** is located between the first side **1001** and the second side **1002**. The third corner **2003** is located between the second side **1002** and the third side **1003**. The fourth corner **2004** is located between the third side **1003** and the fourth side **1004**.

[0029] The optical element drive mechanism **100** includes an immovable part **200**, a first movable part **310**, a second movable part **320**, an elastic assembly **400**, a first drive assembly **500**, a second drive assembly **600**, a guiding assembly **700**, a position sensing assembly **800**, and a circuit assembly **900**.

[0030] Both the first movable part **310** and the second movable part **320** are able to move relative to the immovable part **200**. The elastic assembly **400** is connected to the first movable part **310** and the immovable part **200**, and the elastic assembly **400** is connected to the second movable part **320** and the immovable part **200**. The first drive assembly **500** can drive the first movable part **310** to move relative to the immovable part **200** along the central axis C, focusing on the object to be captured, thereby achieving auto focus (AF). The second drive assembly **600** can drive the second movable part **320** to move relative to the immovable part **200** along the first axis A1 and the second axis A2, compensating for displacement caused by user-induced shaking or external impact, thereby achieving optical image stabilization (OIS). The guiding assembly **700** may guide the movement of the first movable part **310** and the movement of the second movable part **320**. The position sensing assembly **800** can detect the movement of the first movable part **310**. The circuit assembly **900** may transmit current.

[0031] The immovable part **200** includes a casing **210**, a mounting element **220**, and a bottom **230**. The elastic assembly **400** includes a plurality of first elastic elements **410**, two pop-up elastic elements **420**, and two second elastic elements **430**. The first drive assembly **500** includes a first-side coil **510**, a first-side magnetic element **520**, a third-side coil **530**, and a third-side magnetic element **540**. The second drive assembly **600** includes a plurality of drive elements **610**.

[0032] The guiding assembly **700** includes a second-corner guiding element **710**, a second-corner magnetic element **720**, a fourth-corner guiding element **730**, a fourth-corner magnetic element **740**, a plurality of guiding spherical elements **750**, and a plurality of guiding magnetic elements **760**. The position sensing assembly **800** includes a position sensing element **810**. The circuit assembly **900** includes a first-side circuit element **910**, a third-side circuit element **920**, and an external connection circuit element **930**. The description is merely an example. The elements can be added or omitted according to actual needs.

[0033] A first optical element (not shown), a second optical element (not shown), and a third optical element (not shown) can cooperate with the optical element drive mechanism **100**. The first optical element may be disposed in the first movable part **310**, and the movement of the first movable part **310** may drive the first optical element to move. The first optical element may be one or more lenses, which may be made of plastic or glass.

[0034] The second optical element may be mounted to the second movable part **320**, and the movement of the second movable part **320** may drive the second optical element to move. The second optical element may be an image sensor, such as a charge-coupled device (CCD) or a complementary metal oxide semiconductor (CMOS) image sensor (CIS). Light may be imaged onto the second optical element. The third optical element can be placed over the optical element drive mechanism **100**. The third optical element may be an aperture that controls the amount of light entering the optical element drive mechanism **100**.

[0035] Next, in addition to FIG. **1** and FIG. **2**, please refer to FIG. **3** to FIG. **5** to understand the immovable part **200**. FIG. **3** is a schematic view of the casing **210** in accordance with some embodiments. FIG. **4** is a schematic view of the mounting element **220** in accordance with some embodiments. FIG. **5** is a schematic view of the bottom **230** in accordance with some embodiments.

[0036] The casing **210** may accommodate the bottom **230**, the first movable part **310**, the elastic assembly **400**, the first drive assembly **500**, etc. The casing **210** may be made of plastic or metal. The casing **210** includes a casing upper portion **211** and a casing lower portion **212**. The casing upper portion **211** may have a circular or polygonal profile. The casing lower portion **212** may have a polygonal profile, such as quadrilateral. From a top-down view, the casing upper portion **211** is positioned within the boundaries of the casing lower portion **212**.

[0037] The casing upper portion **211** includes two casing recesses **2111**. The two casing recesses **2111** are formed near the first corner **2001** and the third corner **2003**, respectively. The casing lower portion **212** includes a top wall **2121** and a plurality of side walls **2122**. The top wall **2121** is perpendicular to the central axis C. The side walls **2122** extend along the central axis C from the edges of the top wall **2121**.

[0038] The mounting element **220** is disposed between the casing **210** and the first movable part **310**. The mounting element **220** may at least partially cover the top surface of the first movable part **310** and the top surface of the bottom **230**. In some embodiments, an adhesive (such as glue) may be applied to the surface of the mounting element **220** that faces the bottom **230** (for example, a plurality of inner surfaces of the mounting element **220**) first, and the mounting element **220** is subsequently disposed on the bottom **230**. The mounting element **220** may include metal. In some embodiments, the mounting element **220** is manufactured using stamping processes.

[0039] In some embodiments, the surfaces of casing **210** and the mounting element **220** that face each other (for example, the inner surface of the casing upper portion **211** and the top surface of the mounting element **220**) include an anti-reflective material to reduce stray light. The mounting element **220** includes two mounting-element holes **221**. The two mounting-element holes **221** are formed close to the second corner **2002** and the fourth corner **2004**, respectively.

[0040] The bottom **230** is disposed between the first movable part **310** and the second movable part **320**. The bottom **230** may accommodate the first movable part **310**. The bottom **230** includes a bottom upper portion **231** and a bottom lower portion **232**. The bottom upper portion **231** may have a circular or polygonal profile, which may be similar to the profile or shape of the first movable part **310**. The bottom lower portion **232** may have a polygonal profile, such as quadrilateral. From a top-down view, the bottom upper portion **231** is positioned within the boundaries of the bottom lower portion **232**.

[0041] The bottom **230** may further include an embedded circuit **233** embedded therein. In some embodiments, the embedded circuit **233** may be formed in the bottom **230** by insert molding. The embedded circuit **233** may be made of a conductive material, such as metal. For ease of explanation, the embedded circuit **233** is illustrated separately in FIG. **5**. It should be noted that the embedded circuit **233** is actually embedded in the bottom **230**.

[0042] Next, in addition to FIG. **1** and FIG. **2**, please also refer to FIG. **6** and FIG. **7** to understand the first movable part **310** and the second movable part **320**. FIG. **6** is a perspective view of the first movable part **310** in accordance with some embodiments. FIG. **7** is a perspective view of the

second movable part **320** in accordance with some embodiments.

[0043] The first movable part **310** includes a first-movable-part opening **311** and two first-movable-part recesses **312**. The first-movable-part opening **311** can house the first optical element. The two first-movable-part recesses **312** are formed near the second corner **2002** and the fourth corner **2004**, respectively. Only one first-movable-part recess **312** is shown in FIG. **6** due to the perspective. The portions of the first movable part **310** near the first side **1001** and the third side **1003** have a straight-line segment, and the portions of the first movable part **310** near the second side **1002** and the fourth side **1004** have a curved-line segment.

[0044] The second movable part **320** includes a second-movable-part opening **321** and two second-movable-part pillars **322**. The second-movable-part opening **321** can house the second optical element. The two second-movable-part pillars **322** are positioned near the first corner **2001** and the third corner **2003**, respectively.

[0045] Please refer to FIG. **2** to understand the elastic assembly **400**. The first elastic elements **410**, the pop-up elastic elements **420**, and the second elastic elements **430** are made of elastic materials or flexible materials, such as metal. In the technical field, the first elastic elements **410**, the pop-up elastic elements **420**, and the second elastic elements **430** may be referred to as “springs” or “plate-like springs.”

[0046] In some embodiments, there are four first elastic elements **410**, grouped in two pairs. For example, one pair of two first elastic elements **410** is positioned near the second side **1002**, and the other pair of two first elastic elements **410** is positioned near the fourth side **1004**.

[0047] Each first elastic element **410** includes a portion connected to the bottom **230** (not labeled), a portion connected to the first movable part **310** (not labeled), and a deformable portion therebetween. Similarly, each second elastic element **430** includes a portion connected to the bottom **230** (not labeled), a portion connected to the second movable part **320** (not labeled), and a deformable portion therebetween.

[0048] The first elastic elements **410** and the second elastic elements **430** can deform due to the extension or shrinkage of their deformation portions. Due to the first elastic elements **410** and the second elastic elements **430**, when the first movable part **310** and/or the second movable part **320** are driven to move relative to the immovable part **200**, the range of motion of the first movable part **310** and/or the second movable part **320** is restricted. Therefore, the first movable part **310** and/or the second movable part **320** are prevented from being damaged because of collision with other elements when the optical element drive mechanism **100** moves or is impacted.

[0049] The pop-up elastic elements **420** are disposed between the first movable part **310** and the second movable part **320**. The two pop-up elastic elements **420** are disposed near the second side **1002** and the fourth side **1004**, respectively.

[0050] Next, please still refer to FIG. **2** to understand the first drive assembly **500**. The first-side coil **510** and the third-side coil **530** may include or may be a coil wound with enameled wire, FP coils, etc., and may be a combination with a flexible printed circuit board (FPC), a flex-rigid composite board, etc. In some embodiments, both the first-side coil **510** and the third-side coil **530** are a combination of FP coils and an FPC.

[0051] The first-side magnetic element **520** and the third-side magnetic element **540** may be magnets. In some embodiments, the first-side magnetic element **520** and the third-side magnetic element **540** may be magnetic elements arranged in a Halbach magnetic array.

[0052] The first-side coil **510** and the first-side magnetic element **520** are disposed on the first side **1001**. The position of the first-side coil **510** corresponds to the position of the first-side magnetic element **520**. The third-side coil **530** and the third-side magnetic element **540** are disposed on the third side **1003**. The position of the third-side coil **530** corresponds to the position of the third-side magnetic element **540**.

[0053] The driving force (magnetic force) generated between the first-side coil **510** and the first-side magnetic element **520** and the driving force (magnetic force) generated between the third-side

coil **530** and the third-side magnetic element **540** can drive the first movable part **310** together with the first optical element to move relative to the immovable part **200** along the central axis C.

[0054] Since the first-side magnetic element **520**, the third-side magnetic element **540**, and the two pop-up elastic elements **420** are respectively disposed on different sides of the optical element drive mechanism **100**, space can be effectively utilized and miniaturization can be achieved.

[0055] Next, in addition to FIG. **1** and FIG. **2**, please also refer to FIG. **8A**, FIG. **8B**, and FIG. **8C** to understand the second drive assembly **600**. FIG. **8A**, FIG. **8B**, and FIG. **8C** are schematic views illustrating the operation of the second drive assembly **600** in driving the second movable part **320** in accordance with some embodiments.

[0056] The drive elements **610** are disposed below the second movable part **320**. In some embodiments, the second drive assembly **600** includes four drive elements **610** disposed on the first side **1001**, the second side **1002**, the third side **1003**, and the fourth side **1004**, respectively. In some embodiments, the drive elements **610** are made of a shape memory alloy (SMA), for example, an iron-based alloy, a copper-based alloy (for example, a copper-zinc-aluminum alloy, a copper-aluminum-nickel alloy), a titanium-nickel alloy, a titanium-palladium alloy, a titanium-nickel-copper alloy, a titanium-nickel-palladium alloy, a gold-cadmium alloy, an indium-thallium alloy, or a combination thereof.

[0057] The shape memory alloy deforms when the temperature changes. Therefore, the same or different driving signal may be applied to the four drive elements **610** to independently control the temperature of each drive element **610**, thereby changing the length of each drive element **610**. For example, when a driving signal is applied to change the temperature of the drive elements **610**, the lengths of the drive elements **610** are extended or shortened, thereby driving the second movable part **320** and the second optical element on the second movable part **320** to move. As a result, the positional relationship of the second movable part **320** relative to the immovable part **200** is altered. When the driving signal is no longer applied, each drive element **610** returns to its original length.

[0058] It should be noted that two ends of each of the drive elements **610** are respectively connected to the bottom **230** and the second movable part **320**. When the lengths of the drive elements **610** change, the bottom **230** remains stationary. For ease of explanation, FIG. **8A** to FIG. **8C** illustrate the bottom **230** and the second movable part **320** in a rather simplified manner, and the four drive elements **610** are further defined as a first drive element **610A**, a second drive element **610B**, a third drive element **610C**, and a fourth drive element **610D**.

[0059] As shown in FIG. **8A**, at this time, no driving signal is applied, and the first drive element **610A**, the second drive element **610B**, the third drive element **610C**, and the fourth drive element **610D** all maintain their original lengths and are symmetrically arranged in pairs.

[0060] As shown in FIG. **8B**, when the applied driving signal causes the second drive element **610B** to extend and the fourth drive element **610D** to shorten, the second movable part **320** can perform position correction and displacement compensation relative to the immovable part **200** in the direction indicated by the arrow (negative X-axis). It can be inferred from FIG. **8B** that when the length of the second drive element **610B** is shortened and the length of the fourth drive element **610D** is lengthened, the second movable part **320** can perform position correction and displacement compensation relative to the immovable part **200** toward the positive X-axis.

[0061] As shown in FIG. **8C**, when the applied driving signal causes the length of the first drive element **610A** to lengthen and the length of the third drive element **610C** to shorten, the second movable part **320** can perform position correction and displacement compensation relative to the immovable part **200** in the direction indicated by the arrow (positive Y-axis). It can be inferred from FIG. **8C** that when the length of the first drive element **610A** is extended and the length of the third drive element **610C** is shortened, the second movable part **320** can perform position correction and displacement compensation relative to the immovable part **200** toward the negative Y-axis.

[0062] In summary, by applying appropriate driving signals to the first drive assembly **500** and/or the second drive assembly **600**, the first drive assembly **500** can drive the first movable part **310** to move relative to the immovable part **200** along a direction parallel with the central axis C, and the second drive assembly **600** can drive the second movable part **320** to move relative to the immovable part **200** along a direction perpendicular to the central axis C. In this technical field, the movement of the second optical element, caused by the motion of the second movable part **320** with respect to the immovable part **200** along a direction perpendicular to the central axis C, is referred to as sensor shift.

[0063] The image quality produced by the optical element drive mechanism **100** can be enhanced through the movement of the first movable part **310** and/or the second movable part **320**. Additionally, the first drive assembly **500** and the second drive assembly **600** can generate relatively large driving forces, enabling the optical element drive mechanism **100** to support larger or heavier optical elements. In some embodiments, the second optical element may be a one-inch image sensor.

[0064] Next, please refer to FIG. 2 alone to understand the guiding assembly **700** and the position sensing assembly **800**.

[0065] The second-corner guiding element **710** and the fourth-corner guiding element **730** may have substantially the same structure. The second-corner guiding element **710** and the fourth-corner guiding element **730** may have a long strip structure extending along the central axis C, such as rod-like or a bar-like. The second-corner guiding element **710** and the fourth-corner guiding element **730** may include metal or plastic.

[0066] The second-corner magnetic element **720** and the fourth-corner magnetic element **740** may have substantially the same structure. The second-corner magnetic element **720** and the fourth-corner magnetic element **740** may be polygonal. The second-corner magnetic element **720** and the fourth-corner magnetic element **740** may be magnetically-permeable elements or magnets. A magnetically-permeable element refers to an element including a material with high magnetic permeability, such as ferromagnetic materials, including iron (Fe), nickel (Ni), cobalt (Co), or an alloy thereof.

[0067] The second-corner guiding element **710** and the second-corner magnetic element **720** are disposed near the second corner **2002**. The fourth-corner guiding element **730** and the fourth-corner magnetic element **740** are disposed near the fourth corner **2004**. In some embodiments, the second-corner magnetic element **720** and the fourth-corner magnetic element **740** are respectively disposed in the two first-movable-part recesses **312** of the first movable part **310**.

[0068] The second-corner magnetic element **720** can generate magnetic attraction force with the first-side magnetic element **520**, and the fourth-corner magnetic element **740** can generate magnetic attraction with the third-side magnetic element **540**, so that the second-corner guiding element **710** and the fourth-corner guiding element **730** are in close contact with the first movable part **310**. As a result, it is ensured that the first movable part **310** moves relative to the immovable part **200** along the central axis C, and the possibility of the first movable part **310** generating unwanted shaking is reduced.

[0069] In some embodiments, the top of the second-corner guiding element **710** and the top of the fourth-corner guiding element **730** can be respectively connected to the two mounting-element holes **221** (for example, by welding) to reduce the possibility of the second-corner guiding element **710** and the fourth-corner guiding element **730** falling off from the optical element drive mechanism **100**.

[0070] The guiding spherical elements **750** and the guiding magnetic elements **760** are disposed between the bottom **230** and the second movable part **320**. The guiding spherical elements **750** and the guiding magnetic elements **760** can ensure the second movable part **320** to move relative to the immovable part **200** on a plane perpendicular to the central axis C.

[0071] When the first movable part **310** moves relative to the immovable part **200**, the position of

the first movable part **310** can be detected by the position sensing assembly **800**. The position sensing element **810** may be disposed near the first corner **2001**. In some embodiments, the position sensing element **810** may be a Hall sensing element, a Giant Magneto Resistance (GMR) sensing element, a Tunneling Magneto Resistance (TMR) sensing element, etc.

[0072] The position sensing element **810** obtains the position of the first movable part **310** by sensing the change of magnetic lines of force (including but not limited to the density and direction of magnetic lines of force) of the first-side magnetic element **520**. Since the second-corner guiding element **710** for guiding the first movable part **310**, the fourth-corner guiding element **730** for guiding the first movable part **310**, and the position sensing element **810** for sensing the first movable part **310** are disposed at different corners of the optical element drive mechanism **100**, space can be effectively utilized and miniaturization can be achieved.

[0073] It should be noted that the first-side magnetic element **520** is a drive magnetic element of the first drive assembly **500** and a reference magnetic element of the position sensing assembly **800**. That is, a single magnetic element can have multiple functions, so the volume of the optical element drive mechanism **100** can be reduced to further achieve miniaturization.

[0074] Next, in addition to FIG. 2, please also refer to FIG. 9, FIG. 10, FIG. 11, and FIG. 12 to understand the circuit assembly **900**. FIG. 9 is a schematic view of the first-side circuit element **910** and the third-side circuit element **920** in accordance with some embodiments. FIG. 10 is a schematic view of the external connection circuit element **930** in accordance with some embodiments. FIG. 11 and FIG. 12 are schematic views of the bottom **230** and the circuit assembly **900** from different perspectives in accordance with some embodiments.

[0075] The first-side circuit element **910** is disposed on the first side **1001**. The first-side circuit element **910** may be an FPC. The first-side circuit element **910** is electrically connected to the first drive assembly **500** and the embedded circuit **233** embedded in the bottom **230**. The first-side circuit element **910** includes a first portion of the first-side circuit element **911**, a second portion of the first-side circuit element **912**, a bent portion of the first-side circuit element **913**, and a protruding portion of the first-side circuit element **914**.

[0076] The first portion of the first-side circuit element **911** has a plate-like structure. The first portion of the first-side circuit element **911** is perpendicular to the central axis C. The first portion of the first-side circuit element **911** is disposed on the bottom **230**. For example, the first portion of the first-side circuit element **911** may directly contact a bottom first surface **235** (only labeled in FIG. 12) of the bottom **230** that faces the first portion of the first-side circuit element **911**.

[0077] Specifically, the bottom first surface **235** faces the second movable part **320**. In some embodiments, when viewed along a direction that is perpendicular to the central axis C, the bottom first surface **235** is located between the center of the first movable part **310** and the bottom surface of the first portion of the first-side circuit element **911**. In other words, the center of the first movable part **310** is closer to the top wall **2121** of the casing **210** than the bottom first surface **235**, and the bottom surface of the first portion of the first-side circuit element **911** is closer to the second movable part **320** than the bottom first surface **235**.

[0078] The second portion of the first-side circuit element **912** has a plate-like structure. The second portion of the first-side circuit element **912** is substantially parallel with the central axis C. That is, the second portion of the first-side circuit element **912** is not parallel with the first portion of the first-side circuit element **911**. The second portion of the first-side circuit element **912** is disposed on the bottom **230**. For example, the second portion of the first-side circuit element **912** may directly contact a bottom side surface (not shown) of the bottom **230** that faces the second portion of the first-side circuit element **912**. Specifically, the bottom side surface faces the first movable part **310**.

[0079] The first drive assembly **500** is electrically connected to the first portion of the first-side circuit element **911** via the second portion of the first-side circuit element **912**. That is, the current flowing out from the first drive assembly **500** first flows through the second portion of the first-side

circuit element **912** and then flows into the first portion of the first-side circuit element **911**.

[0080] The first-side coil **510** includes an outer surface of the first-side coil **515** (only labeled in FIG. 2) facing the second portion of the first-side circuit element **912**. When viewed along a direction that is perpendicular to the center axis C, the outer surface of the first-side coil **515** at least partially overlaps the first portion of the first-side circuit element **911**. Furthermore, when viewed along a direction that is perpendicular to the central axis C, the center of the outer surface of the first-side coil **515** is located between the position sensing element **810** and the second-corner guiding element **710**. In other words, the position sensing element **810** is closer to the first corner **2001** than the center of the outer surface of the first-side coil **515**, and the second-corner guiding element **710** is closer to the second corner **2002** than the center of the outer surface of the first-side coil **515**.

[0081] The bent portion of the first-side circuit element **913** may be connected to the first portion of the first-side circuit element **911** and the second portion of the first-side circuit element **912**.

[0082] The protruding portion of the first-side circuit element **914** has a plate-like structure. The protruding portion of the first-side circuit element **914** is not parallel with the first portion of the first-side circuit element **911**, and the protruding portion of the first-side circuit element **914** is parallel with the second portion of the first-side circuit element **912**. When viewed along a direction that is perpendicular to the central axis C, the protruding portion of the first-side circuit element **914** does not overlap the second portion of the first-side circuit element **912**. In some embodiments, the position sensing element **810** may be disposed on the protruding portion of the first-side circuit element **914**. Therefore, the position sensing assembly **800** may be electrically connected to the first portion of the first-side circuit element **911** via the protruding portion of the first-side circuit element **914**.

[0083] The third-side circuit element **920** is disposed on the third side **1003**. The third-side circuit element **920** may be an FPC. The third-side circuit element **920** includes a first portion of the third-side circuit element **921**, a second portion of the third-side circuit element **922**, and a bent portion of the third-side circuit element **923** similar to the first portion of the first-side circuit element **911**, the second portion of the first-side circuit element **912**, and the bent portion of the first-side circuit element **913**.

[0084] The primary difference between the third-side circuit element **920** and the first-side circuit element **910** is that the third-side circuit element **920** may lack a protruding portion. This is because, in the present disclosure, only a single position sensing element (e.g., the position sensing element **810** described above) is required to detect the movement of the first movable part **310** relative to the immovable part **200** along the central axis C. Consequently, only one protruding portion of the circuit element is necessary for the placement of the single position sensing element.

[0085] The first-side circuit element **910** is electrically connected to an external circuit via the external connection circuit element **930**. The external connection circuit element **930** includes a first portion of the external connection circuit element **931**, a second portion of the external connection circuit element **932**, a third portion of the external connection circuit element **933**, and a movable portion of the external connection circuit element **934**.

[0086] The first portion of the external connection circuit element **931** has a plate-like structure. The first portion of the external connection circuit element **931** is parallel with the central axis C. The first portion of the external connection circuit element **931** is disposed on the first side **1001**. The first portion of the external connection circuit element **931** may have a plurality of pins **9311** electrically connected to the external circuit. Since the pins **9311** and the first-side magnetic element **520** are located on the same side, the optical element drive mechanism **100** can be conveniently positioned alongside other optical element drive mechanisms and/or optical modules within an electronic device, while minimizing magnetic interference.

[0087] The second portion of the external connection circuit element **932** has a plate-like structure. The second portion of the external connection circuit element **932** is perpendicular to the central

axis C. The second portion of the external connection circuit element **932** is not parallel with the first portion of the external connection circuit element **931**. In some embodiments, the second portion of the external connection circuit element **932** is perpendicular to the first portion of the external connection circuit element **931**. In some embodiments, the second portion of the external connection circuit element **932** and the first portion of the first-side circuit element **911** are parallel with each other.

[0088] The second portion of the external connection circuit element **932** is electrically connected to the external circuit via the first portion of the external connection circuit element **931**. That is, the current flowing in from the external circuit first flows through the first portion of the external connection circuit element **931** and then flows into the second portion of the external connection circuit element **932**.

[0089] The second portion of the external connection circuit element **932** is disposed on the first side **1001**. The second portion of the external connection circuit element **932** is disposed on the bottom **230**. For example, the second portion of the external connection circuit element **932** may directly contact a bottom second surface **236** (only labeled in FIG. **11**) of the bottom **230** that faces the second portion of the external connection circuit element **932**.

[0090] Specifically, the bottom second surface **236** faces the top wall **2121** of the casing **210**. In some embodiments, when viewed along a direction that is perpendicular to the central axis C, the bottom second surface **236** is located between the center of the first movable part **310** and the bottom first surface **235**. In other words, the center of the first movable part **310** is closer to the top wall **2121** of the casing **210** than the bottom second surface **236**, and the bottom first surface **235** is closer to the second movable part **320** than the bottom second surface **236**.

[0091] The third portion of the external connection circuit element **933** has a plate-like structure. The third portion of the external connection circuit element **933** is not parallel with the first portion of the external connection circuit element **931** and the second portion of the external connection circuit element **932**. The third portion of the external connection circuit element **933** is disposed on the second side **1002**. The third portion of the external connection circuit element **933** is disposed on the second movable part **320**. For example, the third portion of the external connection circuit element **933** may directly contact an outer surface (not labeled) of the second movable part **320** that faces the third portion of the external connection circuit element **933**.

[0092] The movable portion of the external connection circuit element **934** is connected to the first portion of the external connection circuit element **931** and the third portion of the external connection circuit element **933**. In some embodiments, the movable portion of the external connection circuit element **934** surrounds the second corner **2002**. In some embodiments, the movable portion of the external connection circuit element **934** may not contact any other element. That is, the movable portion of the external connection circuit element **934** is suspended in mid-air.

[0093] In some embodiments, when viewed along a direction that is perpendicular to the central axis C, the bottom first surface **235** and the bottom second surface **236** are located between the second portion of the external connection circuit element **932** and the first portion of the first-side circuit element **911**. In other words, the bottom first surface **235** and the bottom second surface **236** are sandwiched between the external connection circuit element **930** and the first-side circuit element **910**.

[0094] Next, please refer to FIG. **13** to FIG. **19** to understand some other features of the optical element drive mechanism **100**, such as the positional relationship between the elements, circuit connection methods, and other details. FIG. **13**, FIG. **14**, and FIG. **15** are perspective views of the optical element drive mechanism **100** with the casing **210** omitted from different perspectives in accordance with some embodiments. FIG. **16** is a top view of the optical element drive mechanism **100** with the casing **210** omitted in accordance with some embodiments. FIG. **17** and FIG. **18** are perspective views of the optical element drive mechanism **100** with the casing **210**, the mounting element **220**, and the second movable part **320** omitted in accordance with some embodiments.

FIG. 19 is a cross-sectional view of the optical element drive mechanism **100** with the casing **210** omitted in accordance with some embodiments.

[0095] The first portion of the first-side circuit element **911** may include an electrical connection portion of the first-side circuit element **9111**. The first portion of the first-side circuit element **911** is electrically connected to the embedded circuit **233** of the bottom **230** via the electrical connection portion of the first-side circuit element **9111**. Specifically, a solder material (e.g., a solder ball) may be applied on the electrical connection portion of the first-side circuit element **9111**, so the electrical connection portion of the first-side circuit element **9111** serves as a soldering point. As a result, the electrical connection of the first-side circuit element **910** is established at the bottom first surface **235** of the bottom **230**.

[0096] The second portion of the external connection circuit element **932** may include an electrical connection portion of the external connection circuit element **9321**. The second portion of the external connection circuit element **932** is electrically connected to the embedded circuit **233** of the bottom **230** via the electrical connection portion of the external connection circuit element **9321**. Specifically, a solder material (e.g., a solder ball) may be applied on the electrical connection portion of the external connection circuit element **9321**, so the electrical connection portion of the external connection circuit element **9321** serves as a soldering point. As a result, the electrical connection of the external connection circuit element **930** is established at the bottom second surface **236** of the bottom **230**.

[0097] In some embodiments, when viewed along a direction that is perpendicular to the central axis C, the electrical connection portion of the first-side circuit element **9111** and the electrical connection portion of the external connection circuit element **9321** do not overlap. That is, the electrical connection portion of the first-side circuit element **9111** and the electrical connection portion of the external connection circuit element **9321** are located at different horizontal planes.

[0098] In summary, the first-side circuit element **910** is electrically connected to the embedded circuit **233** of the bottom **230** and the first drive assembly **500** at the bottom first surface **235** of the bottom **230** via the electrical connection portion of the first-side circuit element **9111**. Similarly, the external connection circuit element **930** is electrically connected to the embedded circuit **233** of the bottom **230** at the bottom second surface **236** via the electrical connection portion of the external connection circuit element **9321**. As a result, the circuit assembly **900** can improve circuit designs. Furthermore, the design of the first-side circuit element **910**, the third-side circuit element **920**, and the external connection circuit element **930** allows for easy assembly of the circuit assembly **900**. For example, the first-side circuit element **910** and/or the third-side circuit element **920** may be mounted from bottom to top.

[0099] Furthermore, as shown in FIG. 16, the end of one of the first elastic elements **410** may have a receiving structure for receiving the end of another first elastic element **410**, which can save materials and reduce waste. In some embodiments, a portion or all of the first elastic element **410** may be exposed to facilitate electrical connection or physical connection with the third optical element.

[0100] As shown in FIG. 19, the presence of the pop-up elastic elements **420** divides the movement of the first movable part **310** into a focusing section and a non-focusing section. When the first movable part **310** compresses the pop-up elastic elements **420**, its movement corresponds to the non-focusing section.

[0101] In summary, the present disclosure provides an optical element drive mechanism that drives different optical elements to move by using different driving assemblies. These drive assemblies can generate relatively large driving forces, enabling the optical element drive mechanism to support larger or heavier optical elements. The circuit assembly includes a plurality of circuit elements. The circuit assembly can improve circuit designs. Furthermore, the design of the circuit assembly allows for easy assembly. Besides, the elastic assembly include pop-up elastic elements. Furthermore, the arrangement of assemblies can effectively utilize space and achieve

miniaturization.

[0102] The foregoing outlines features of several embodiments, so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced in the following description. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations in the following description without departing from the spirit and scope of the present disclosure.

Claims

1. An optical element drive mechanism having a central axis, comprising: an immovable part comprising a bottom; a first movable part connected to a first optical element, wherein the first movable part is movable relative to the bottom; and a first drive assembly driving the first movable part to move relative to the bottom.
2. The optical element drive mechanism as claimed in claim 1, further comprising a first-side circuit element electrically connected to the first drive assembly, wherein the bottom comprises an embedded circuit, and the first-side circuit element comprises: a first portion of the first-side circuit element having a plate-like structure, wherein the first portion of the first-side circuit element is disposed on the bottom, wherein the first portion of the first-side circuit element comprises an electrical connection portion of the first-side circuit element, and the first portion of the first-side circuit element is electrically connected to the embedded circuit of the bottom via the electrical connection portion of the first-side circuit element, and wherein the bottom comprises a bottom first surface facing the first portion of the first-side circuit element, and when viewed along a direction that is perpendicular to the central axis, the bottom first surface is located between a center of the first movable part and the first portion of the first-side circuit element.
3. The optical element drive mechanism as claimed in claim 2, wherein the first-side circuit element further comprises: a second portion of the first-side circuit element having a plate-like structure and not parallel with the first portion of the first-side circuit element; and a bent portion of the first-side circuit element connected to the first portion of the first-side circuit element and the second portion of the first-side circuit element, wherein the first drive assembly is electrically connected to the first portion of the first-side circuit element via the second portion of the first-side circuit element, and wherein the first drive assembly comprises a first-side coil, the first-side coil has an outer surface of the first-side coil facing the second portion of the first-side circuit element, and when viewed along a direction that is perpendicular to the central axis, the outer surface of the first-side coil at least partially overlaps the first portion of the first-side circuit element.
4. The optical element drive mechanism as claimed in claim 3, wherein the first-side circuit element further comprises: a protruding portion of the first-side circuit element having a plate-like structure, not parallel with the first portion of the first-side circuit element, and parallel with the second portion of the first-side circuit element, wherein when viewed along a direction that is perpendicular to the central axis, the protruding portion of the first-side circuit element does not overlap the second portion of the first-side circuit element.
5. The optical element drive mechanism as claimed in claim 4, further comprising a position sensing assembly sensing the movement of the first movable part, wherein the position sensing assembly comprises a position sensing element disposed on the protruding portion of the first-side circuit element, and the position sensing assembly is electrically connected to the first portion of the first-side circuit element via the protruding portion of the first-side circuit element.
6. The optical element drive mechanism as claimed in claim 3, further comprising a guiding assembly guiding the movement of the first movable part relative to the bottom and a position

sensing element sensing the movement of the first movable part, wherein the guiding assembly comprises a second-corner guiding element, and when viewed along a direction that is perpendicular to the central axis, a center of the outer surface of the first-side coil is located between the position sensing element and the second-corner guiding element.

7. The optical element drive mechanism as claimed in claim 2, further comprising an external connection circuit element, wherein the first-side circuit element is electrically connected to an external circuit via the external connection circuit element, and the external connection circuit element comprises: a first portion of the external connection circuit element having a plate-like structure; and a second portion of the external connection circuit element having a plate-like structure and not parallel with the first portion of the external connection circuit element, wherein the second portion of the external connection circuit element comprises an electrical connection portion of the external connection circuit element, and the second portion of the external connection circuit element is electrically connected to the embedded circuit of the bottom via the electrical connection portion of the external connection circuit element, wherein the second portion of the external connection circuit element is electrically connected to the external circuit via the first portion of the external connection circuit element, and wherein the second portion of the external connection circuit element is parallel with the first portion of the first-side circuit element.

8. The optical element drive mechanism as claimed in claim 7, wherein when viewed along a direction that is perpendicular to the central axis, the bottom first surface is located between the first portion of the first-side circuit element and the second portion of the external connection circuit element, and when viewed along a direction that is perpendicular to the central axis, the electrical connection portion of the first-side circuit element does not overlap the electrical connection portion of the external connection circuit element.

9. The optical element drive mechanism as claimed in claim 7, further comprising a second movable part connected to a second optical element, wherein the external connection circuit element further comprises: a third portion of the external connection circuit element having a plate-like structure and neither parallel with the first portion of the external connection circuit element and the second portion of the external connection circuit element; and a movable portion of the external connection circuit element connected to the third portion of the external connection circuit element, wherein the bottom is disposed between the first movable part and the second movable part, and the third portion of the external connection circuit element is disposed on the second movable part.

10. The optical element drive mechanism as claimed in claim 1, wherein the immovable part further comprises a casing and a mounting element disposed between the casing and the first movable part, wherein the mounting element at least partially covers a top surface of the first movable part and a top surface of the bottom.

11. The optical element drive mechanism as claimed in claim 10, wherein both the casing and the mounting element comprise an anti-reflective material.

12. The optical element drive mechanism as claimed in claim 10, wherein the mounting element comprises two mounting-element holes located at diagonal corners.

13. The optical element drive mechanism as claimed in claim 12, further comprising a guiding assembly guiding the movement of the first movable part relative to the bottom, wherein the guiding assembly comprises a second-corner guiding element and a fourth-corner guiding element, and a top of the second-corner guiding element and a top of the fourth-corner guiding element are connected to the two mounting-element holes.

14. The optical element drive mechanism as claimed in claim 13, wherein the first movable part comprises two first-movable-part recesses, and the guiding assembly further comprises a second-corner magnetic element and a fourth-corner magnetic element respectively disposed in the two first-movable-part recesses.

15. The optical element drive mechanism as claimed in claim 1, wherein the optical element drive

mechanism includes a first side, a second side, a third side opposite to the first side, and a fourth side opposite to the second side, wherein portions of the first movable part near the first side and the third side have a straight-line segment, and portions of the first movable part near the second side and the fourth side have a curved-line segment.

16. The optical element drive mechanism as claimed in claim 15, further comprising an elastic assembly, wherein the elastic assembly comprises two pop-up elastic elements respectively disposed on the second side and the fourth side.

17. The optical element drive mechanism as claimed in claim 16, further comprising a second movable part disposed below the bottom, wherein the second movable part is movable relative to the bottom, and the pop-up elastic elements are located between the first movable part and the second movable part.

18. The optical element drive mechanism as claimed in claim 1, wherein the first drive assembly comprises a first-side coil, and the first-side coil is a combination of FP coils and an FPC.

19. The optical element drive mechanism as claimed in claim 1, further comprising a circuit assembly electrically connected to the first drive assembly, wherein the circuit assembly comprises: a first-side circuit element comprising a first portion of the first-side circuit element that is perpendicular to the central axis; and an external connection circuit element comprising a first portion of the external connection circuit element that is parallel with the central axis and a second portion of the external connection circuit element that is perpendicular to the central axis, wherein part of the bottom is sandwiched between the first portion of the first-side circuit element and the second portion of the external connection circuit element.

20. The optical element drive mechanism as claimed in claim 19, wherein the external connection circuit element further comprises: a third portion of the external connection circuit element that is parallel with the central axis; and a movable portion of the external connection circuit element that is connected to the first portion of the external connection circuit element and the third portion of the external connection circuit element, wherein the third portion of the external connection circuit element is located on a side of the optical element drive mechanism that is different from another side where the first portion of the external connection circuit element and the second portion of the external connection circuit element are located.
